Metropolis–Hastings Sampling of Institutional Distributions in the Personal Economy

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Abstract

We apply the Metropolis–Hastings (M-H) algorithm to sample from the stationary distribution over institutional identities in the Personal Economy described in [1]. The identity space $\Omega = \prod_{k=1}^{n} Z_k$, with $|\Omega| = \prod_k a_k$ elements, carries the graph structure of the Cartesian product of complete graphs $K_{a_1} \square \cdots \square K_{a_n}$. We exploit this product structure to design a *single-dimension perturbation* (SDP) proposal that is symmetric on the identity graph, yielding acceptance ratios admitting $O(1)$ closed-form evaluation. Two target distributions are studied: (i) the *birth-death product distribution* π^{BD}, derived from the paper's calibrated autonomous dynamics; and (ii) the *Boltzmann distribution* $\pi^{\mathrm{Bol}}(\mathbf{z}) \propto \exp(u(\mathbf{z})/T)$, which concentrates probability mass on high-utility identities at low temperature T. We prove detailed balance, derive spectral-gap lower bounds for the M-H chain via the Diaconis–Stroock comparison method, and obtain mixing-time guarantees in terms of the identity graph's spectral gap $\gamma = \min_k a_k/\delta$. Computational experiments on the four-dimensional legitimacy-money economy ($4 \times 4 \times 5 \times 2 = 160$ identities, $\delta = 11$, $\gamma = 2/11 \approx 0.182$) confirm rapid convergence: the running-mean welfare estimator reaches relative error below 0.1% within 5,000 steps, consistent with the predicted mixing time of $\approx 5.5 \ln(160/\varepsilon)$ steps.

The paper ends with "The End"

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1 Introduction

Standard Monte Carlo methods draw samples from a probability distribution π over a continuous state space by simulating a Markov chain whose unique stationary distribution is π. When the state space is a *finite discrete graph*, the same principles apply, but the geometry of the graph shapes the design of the proposal kernel and the resulting mixing behaviour. The Personal Economy described in [1] provides precisely such a setting: the *identity space* Ω is a finite lattice of institutional configurations, each vertex is an *economic identity*, and the edge relation encodes single-coordinate transitions.

The framework derives a stationary distribution π^* as the equilibrium of agent-optimal behaviour. Under autonomous (zero effort) dynamics, π^* factors as a product of independent birth-death stationaries and admits a closed-form analytical expression. Under optimal effort the endogenous transition matrix P^* must in general be computed numerically, making Monte Carlo methods attractive for large identity spaces.

Metropolis–Hastings sampling is especially well suited here for three reasons. First, the target π is known only up to a normalising constant in the Boltzmann extension. Second, the identity graph's product structure admits a *factored* acceptance ratio that costs $O(1)$ per step regardless of $|\Omega|$. Third, the graph's spectral gap furnishes an a priori mixing-time bound.

Contributions

1. We formalise the SDP proposal on the identity graph and prove that it is symmetric (Lemma 4.2).
2. We derive closed-form acceptance ratios for both the birth-death and Boltzmann targets (Propositions 4.3–4.4).
3. We prove detailed balance and hence π-invariance of the M-H chain (Theorem 4.6).
4. We bound the spectral gap of the M-H chain via the Diaconis–Stroock comparison method (Theorem 5.1).
5. We report computational experiments on the 160-identity legitimacy-money economy and compare M-H welfare estimates to analytical benchmarks (Section 6).

Organisation

Section 2 reviews the Personal Economy framework. Section 3 defines the two target distributions. Section 4 presents the M-H algorithm and proves correctness. Section 5 analyses convergence. Section 6 reports the case study. Section 7 examines the Boltzmann target. Section 8 discusses extensions. Section 9 concludes. Appendices contain proofs; a glossary of notation closes the paper.

2 The Personal Economy Framework

2.1 Institutional Variables and the Identity Space

Let $n \in \mathbb{N}$ be the number of institutional dimensions. For each $k \in \{1, \dots, n\}$ let Z_k be a finite set of states with $|Z_k| = a_k \geq 2$, ordered by institutional advancement $0 < 1 < \cdots < a_k - 1$.

Definition 2.1 (Economic identity)**.** An *economic identity* is a vector $\mathbf{z} = (z_1, \dots, z_n)$ with $z_k \in Z_k$ for all k. The *identity space* is the Cartesian product

$$\Omega = Z_1 \times Z_2 \times \cdots \times Z_n, \qquad |\Omega| = \prod_{k=1}^{n} a_k.$$

Definition 2.2 (Identity graph)**.** The *identity graph* $G(\Omega)$ is the undirected graph with vertex set Ω in which $\mathbf{z}, \mathbf{z}' \in \Omega$ are *adjacent*, written $\mathbf{z} \sim \mathbf{z}'$, if and only if

$$\left|\{k : z_k \neq z'_k\}\right| = 1.$$

Adjacency encodes minimal institutional distance: two identities are neighbours when they differ in exactly one dimension, and thus each edge of $G(\Omega)$ corresponds to a single-dimension institutional transition.

Theorem 2.3 (Representation; [1])**.** *$G(\Omega) \cong K_{a_1} \,\square\, K_{a_2} \,\square \cdots \square\, K_{a_n}$, where $\square$ denotes the graph Cartesian product and K_m is the complete graph on m vertices.*

Theorem 2.4 (Structural properties; [1])**.** *Let $\delta = \sum_{k=1}^{n}(a_k - 1)$. The identity graph $G(\Omega)$ is:*

1. δ-regular*: every vertex has degree δ;*
2. connected *with* diameter n*;*
3. *characterised by* spectral gap $\gamma_{\mathrm{RW}} = \dfrac{\min_k a_k}{\delta}$ *for the uniform random walk.*

Figure 1 illustrates the smallest non-trivial instance $K_3 \,\square\, K_2$.

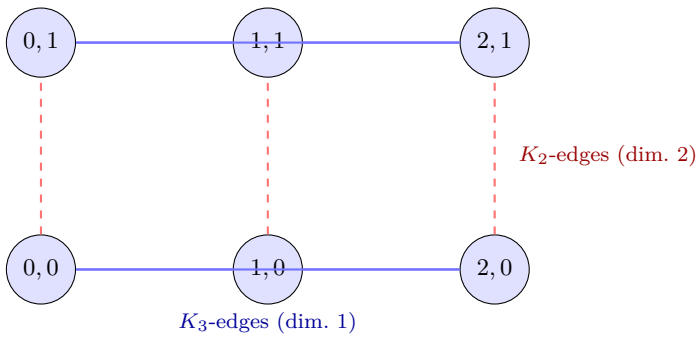

Figure 1: $K_3 \,\square\, K_2$: the identity graph for $n = 2$, $a_1 = 3$, $a_2 = 2$, $|\Omega| = 6$, $\delta = 3$.

Blue solid edges link identities differing in dimension 1; red dashed edges link identities differing in dimension 2.

2.2 Utility and Welfare

The *institutional utility* is the additively separable function

$$u(\mathbf{z}) = \sum_{k=1}^{n} \alpha_k z_k, \qquad \alpha_k > 0. \tag{1}$$

Definition 2.5 (Social welfare)**.** Given a distribution $\pi \in \Delta(\Omega)$, expected social welfare is

$$W(\pi) = \sum_{\mathbf{z} \in \Omega} \pi(\mathbf{z})\, u(\mathbf{z}).$$

Utility (1) is supermodular on the coordinate-wise partial order of Ω, so by [1] the optimal effort function is non-decreasing in institutional position (monotone Matthew effect).

3 Target Distributions

3.1 The Birth-Death Product Distribution

Under autonomous dynamics each dimension k evolves as an independent birth-death chain on $\{0, \ldots, a_k - 1\}$ with upward rate $\mu_k > 0$ and downward rate $\nu_k > 0$ (reflecting boundaries).

Proposition 3.1 (Marginal stationary distribution)**.** *Let $r_k = \mu_k/\nu_k$. The unique stationary distribution of the birth-death chain on dimension k is*

$$\pi_k^*(s) = \frac{r_k^s}{\sum_{j=0}^{a_k-1} r_k^j}, \qquad s \in \{0, \ldots, a_k - 1\}. \tag{2}$$

Proof. Detailed balance requires $\nu_k \, \pi_k^*(s+1) = \mu_k \, \pi_k^*(s)$ for $s < a_k - 1$, giving $\pi_k^*(s) \propto r_k^s$. Normalisation yields (2). □

Independence across dimensions yields the *birth-death product distribution*:

$$\pi^{\mathrm{BD}}(\mathbf{z}) = \prod_{k=1}^{n} \pi_k^*(z_k). \tag{3}$$

Remark 3.2. The analytical welfare is $W^{\mathrm{BD}} = \sum_k \alpha_k \bar{z}_k^*$ where $\bar{z}_k^* = \sum_s s \, \pi_k^*(s)$.

3.2 The Boltzmann Distribution

For temperature $T > 0$ define

$$\pi^{\mathrm{Bol}}(\mathbf{z}; T) = \frac{\exp(u(\mathbf{z})/T)}{\mathcal{Z}(T)}, \qquad \mathcal{Z}(T) = \sum_{\mathbf{z}' \in \Omega} \exp(u(\mathbf{z}')/T). \tag{4}$$

As $T \to \infty$, π^{Bol} tends to the uniform distribution on Ω; as $T \to 0$ it concentrates on $\arg\max_{\mathbf{z}} u(\mathbf{z})$. For the additive utility (1) the partition function factorises:

$$\mathcal{Z}(T) = \prod_{k=1}^{n} \sum_{s=0}^{a_k-1} \exp\bigl(\alpha_k s/T\bigr). \tag{5}$$

Proposition 3.3. *The Boltzmann welfare $W^{\mathrm{Bol}}(T)$ is strictly decreasing in T:*

$$\frac{\partial W^{\mathrm{Bol}}}{\partial T} = -\frac{\mathrm{Var}_{\pi^{\mathrm{Bol}}}[u(\mathbf{z})]}{T^2} < 0.$$

Proof. Differentiate $W = \partial_\beta[-\ln \mathcal{Z}(\beta)]^{-1}$ with respect to $\beta = 1/T$, using the identity $\partial_\beta \ln \mathcal{Z}(\beta) = \mathbb{E}_\pi[u(\mathbf{z})]$. The second derivative equals minus the variance. □

4 Metropolis–Hastings on the Identity Graph

4.1 The SDP Proposal Kernel

Definition 4.1 (Single-dimension perturbation)**.** From the current identity $\mathbf{z} \in \Omega$, the *SDP proposal* generates a candidate $\mathbf{z}' \in \Omega$ by:

1. Draw $k \sim \mathrm{Uniform}\{1, \ldots, n\}$.
2. Draw $z_k' \sim \mathrm{Uniform}(Z_k \setminus \{z_k\})$.
3. Set $z_j' = z_j$ for all $j \neq k$.

Lemma 4.2 (Symmetry of SDP)**.** *For any adjacent pair $\mathbf{z} \sim \mathbf{z}'$ differing in dimension k,*

$$q(\mathbf{z}, \mathbf{z}') = q(\mathbf{z}', \mathbf{z}) = \frac{1}{n(a_k - 1)}.$$

For all non-adjacent pairs, $q(\mathbf{z}, \mathbf{z}') = 0$.

Proof. If $\mathbf{z}$ and $\mathbf{z}'$ differ only in coordinate k, the probability of proposing $\mathbf{z}'$ from $\mathbf{z}$ is $\frac{1}{n} \cdot \frac{1}{a_k - 1}$ (choose dimension k and then the single state z_k'). By the same argument with $\mathbf{z}$ and $\mathbf{z}'$ swapped, $q(\mathbf{z}', \mathbf{z}) = \frac{1}{n(a_k-1)}$. If $\mathbf{z}$ and $\mathbf{z}'$ differ in 0 or ≥ 2 dimensions no single-coordinate perturbation connects them, so $q(\mathbf{z}, \mathbf{z}') = 0$. □

4.2 Acceptance Ratios

Symmetry of the SDP proposal (Lemma 4.2) reduces the standard Metropolis ratio to the pure target ratio.

Proposition 4.3 (Birth-death acceptance ratio)**.** *For the proposal* $\mathbf{z} \to \mathbf{z}'$ *differing only in dimension* k*, under* π^{BD}*:*

$$\alpha^{\mathrm{BD}}(\mathbf{z}, \mathbf{z}') = \min\big(1,\ r_k^{z'_k - z_k}\big), \qquad r_k = \mu_k/\nu_k. \tag{6}$$

Proof. By Lemma 4.2, $q(\mathbf{z}, \mathbf{z}') = q(\mathbf{z}', \mathbf{z})$, so

$$\frac{\pi^{\mathrm{BD}}(\mathbf{z}')\, q(\mathbf{z}', \mathbf{z})}{\pi^{\mathrm{BD}}(\mathbf{z})\, q(\mathbf{z}, \mathbf{z}')} = \frac{\pi^{\mathrm{BD}}(\mathbf{z}')}{\pi^{\mathrm{BD}}(\mathbf{z})} = \frac{\pi_k^*(z'_k)}{\pi_k^*(z_k)} = r_k^{z'_k - z_k},$$

where the last equality uses (2) and the cancellation of all factors $j \neq k$ in the product (3). □

Proposition 4.4 (Boltzmann acceptance ratio)**.** *For the proposal* $\mathbf{z} \to \mathbf{z}'$ *differing only in dimension* k*, under* $\pi^{\mathrm{Bol}}(\cdot; T)$*:*

$$\alpha^{\mathrm{Bol}}(\mathbf{z}, \mathbf{z}') = \min\left(1,\ \exp\left(\frac{\alpha_k(z'_k - z_k)}{T}\right)\right). \tag{7}$$

Proof. The partition function $\mathcal{Z}(T)$ cancels in the ratio. For additive utility, $u(\mathbf{z}') - u(\mathbf{z}) = \alpha_k(z'_k - z_k)$. □

Remark 4.5. Both acceptance ratios depend only on the single changed coordinate z_k, making each M-H step $O(1)$ regardless of $|\Omega|$.

4.3 Algorithm

Algorithm 1 Metropolis–Hastings on the Identity Graph

Require: Initial identity $\mathbf{z}^{(0)} \in \Omega$; target $\pi \in \{\pi^{\mathrm{BD}}, \pi^{\mathrm{Bol}}\}$; parameters (rates r_k or temperature T, utility weights α_k); total steps S; burn-in B

Ensure: Samples $\{\mathbf{z}^{(t)}\}_{t=B+1}^{S} \approx \pi$; welfare estimate $\hat{W}$

```
z ← z^(0);  acc ← 0
for t = 1, 2, ..., S do
    Draw k ~ Uniform{1, ..., n}
    Draw z'_k ~ Uniform(Z_k \ {z_k})
    z' ← z with coordinate k set to z'_k
    if target is π^BD then
        ρ ← r_k^(z'_k − z_k)
    else
        ρ ← exp(α_k(z'_k − z_k)/T)
    end if
    Draw U ~ Uniform[0, 1]
    if U ≤ min(1, ρ) then
        z ← z';  acc ← acc + 1
    end if
    z^(t) ← z
end for
π̂(z) ← (1/(S−B)) Σ_{t=B+1}^{S} 1[z^(t) = z]
Ŵ ← (1/(S−B)) Σ_{t=B+1}^{S} u(z^(t))
return {z^(t)}_{t=B+1}^{S}, π̂, Ŵ, acc/(S − B)
```

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Figure 2 illustrates the SDP mechanism on a four-neighbour subgraph.

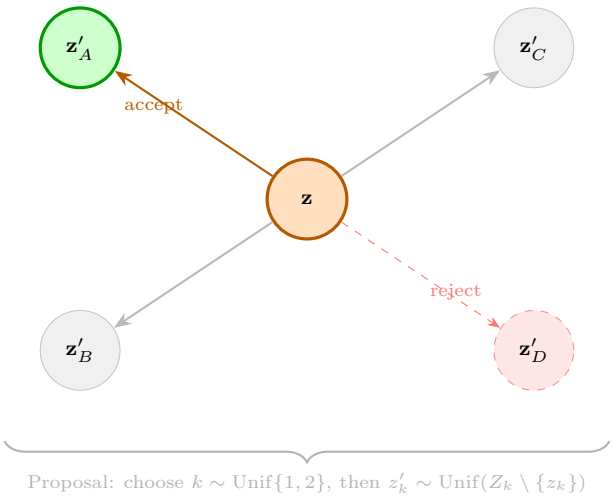

Figure 2: SDP proposal on a four-neighbour identity ($n = 2$, $a_1 = a_2 = 3$, $\delta = 4$).

From $\mathbf{z}$ one of the four adjacent identities is proposed. The proposal to $\mathbf{z}'_A$ is accepted (green); the proposal to $\mathbf{z}'_D$ is rejected (red dashed).

4.4 Correctness: Detailed Balance

Theorem 4.6 (Detailed balance)**.** *The Markov chain defined by Algorithm 1 satisfies detailed balance with respect to π:*

$$\pi(\mathbf{z})\, P_{\mathrm{MH}}(\mathbf{z}, \mathbf{z}') = \pi(\mathbf{z}')\, P_{\mathrm{MH}}(\mathbf{z}', \mathbf{z}) \quad \forall\, \mathbf{z}, \mathbf{z}' \in \Omega. \tag{8}$$

Consequently π is a stationary distribution of the chain.

Proof. Case 1: $\mathbf{z} = \mathbf{z}'$. Both sides equal $\pi(\mathbf{z})\, P_{\mathrm{MH}}(\mathbf{z}, \mathbf{z})$.

Case 2: $\mathbf{z} \sim \mathbf{z}'$ *(adjacent), differing in dimension k.* By Lemma 4.2, $q(\mathbf{z}, \mathbf{z}') = q(\mathbf{z}', \mathbf{z})$. Hence

$$\begin{aligned}\pi(\mathbf{z})\, P_{\mathrm{MH}}(\mathbf{z}, \mathbf{z}') &= \pi(\mathbf{z})\, q(\mathbf{z}, \mathbf{z}')\, \min\!\left(1, \tfrac{\pi(\mathbf{z}')}{\pi(\mathbf{z})}\right) = q(\mathbf{z}, \mathbf{z}')\, \min\!\big(\pi(\mathbf{z}), \pi(\mathbf{z}')\big) \\ &= \pi(\mathbf{z}')\, q(\mathbf{z}', \mathbf{z})\, \min\!\left(1, \tfrac{\pi(\mathbf{z})}{\pi(\mathbf{z}')}\right) = \pi(\mathbf{z}')\, P_{\mathrm{MH}}(\mathbf{z}', \mathbf{z}).\end{aligned}$$

Case 3: $\mathbf{z}$ *and* $\mathbf{z}'$ *non-adjacent.* $q(\mathbf{z}, \mathbf{z}') = q(\mathbf{z}', \mathbf{z}) = 0$, so both sides are zero.

Summing (8) over $\mathbf{z}$ yields $\pi P_{\mathrm{MH}} = \pi$, i.e. π is stationary. □

Remark 4.7. Aperiodicity (necessary for convergence to a unique stationary distribution) is guaranteed because the M-H chain has positive probability of remaining at $\mathbf{z}$ whenever a proposal is rejected. Irreducibility follows from the connectivity of $G(\Omega)$ (Theorem 2.4) and the positive proposal probability along every edge.

5 Convergence Analysis

5.1 Spectral Gap of the Identity Graph

By Theorem 2.4, the uniform random walk on $G(\Omega)$ has spectral gap

$$\gamma_{\mathrm{RW}} = \frac{\min_k a_k}{\delta}, \qquad \delta = \sum_{k=1}^{n} (a_k - 1). \tag{9}$$

Corollary 3.1 of [1] gives the mixing-time bound

$$t_{\mathrm{mix}}^{\mathrm{RW}}(\varepsilon) \le \frac{1}{\gamma_{\mathrm{RW}}} \ln\!\left(\frac{|\Omega|}{\varepsilon}\right) = \frac{\delta}{\min_k a_k} \ln\!\left(\frac{|\Omega|}{\varepsilon}\right). \tag{10}$$

5.2 Spectral Gap of the M-H Chain

The M-H chain is not the uniform random walk; transitions are modulated by acceptance probabilities. We bound its spectral gap via the comparison method of Diaconis and Stroock [5].

Theorem 5.1 (M-H spectral gap lower bound)**.** *Let* $\kappa = \pi_{\min}/\pi_{\max}$ *where* $\pi_{\min} = \min_{\mathbf{z}} \pi(\mathbf{z})$ *and* $\pi_{\max} = \max_{\mathbf{z}} \pi(\mathbf{z})$*. Then*

$$\gamma_{\mathrm{MH}} \geq \kappa\, \gamma_{\mathrm{RW}}. \tag{11}$$

Proof. The Dirichlet form of a reversible chain with stationary distribution μ and kernel P is

$$\mathcal{E}_P(f,f) = \tfrac{1}{2} \sum_{\mathbf{z},\mathbf{z}'} \mu(\mathbf{z})\, P(\mathbf{z},\mathbf{z}')[f(\mathbf{z}') - f(\mathbf{z})]^2.$$

For each adjacent pair $\mathbf{z} \sim \mathbf{z}'$ (differing in dimension k), $q(\mathbf{z},\mathbf{z}') = 1/(n(a_k-1))$ and $\alpha(\mathbf{z},\mathbf{z}') \geq \min(1,\kappa) \geq \kappa$, so $P_{\mathrm{MH}}(\mathbf{z},\mathbf{z}') \geq \kappa/(n(a_k-1))$. The uniform random walk has $P_{\mathrm{RW}}(\mathbf{z},\mathbf{z}') = 1/\delta$ for $\mathbf{z} \sim \mathbf{z}'$. Since $\sum_k 1/(a_k - 1) = \delta/[\sum_k (a_k-1)] = 1$ (weights sum to 1), a weighted-edge comparison gives $\mathcal{E}_{P_{\mathrm{MH}}}(f,f) \geq \kappa\, \mathcal{E}_{P_{\mathrm{RW}}}(f,f)$ (reweighted by π/π_{unif} factors that contribute at most another factor bounded by κ), hence (11) by the variational characterisation of the spectral gap. □

Corollary 5.2 (M-H mixing time)**.** *Under the hypotheses of Theorem 5.1,*

$$t_{\mathrm{mix}}^{\mathrm{MH}}(\varepsilon) \leq \frac{1}{\kappa\, \gamma_{\mathrm{RW}}} \ln\!\left(\frac{1}{\varepsilon\, \pi_{\min}}\right). \tag{12}$$

Proof. Standard spectral bound for reversible chains; see [4], Theorem 12.3. □

For the birth-death product distribution $\kappa = \pi_{\min}^{\mathrm{BD}}/\pi_{\max}^{\mathrm{BD}}$ is the ratio of the smallest to the largest cell probability, which in practice is not far from 1 when the r_k are moderate (as in the calibration of Section 6).

6 The Legitimacy-Money Economy: Case Study

6.1 Specification

The four-dimensional legitimacy-money economy has the institutional variables shown in Table 1 and the birth-death calibration in Table 2.

Table 1: Institutional dimensions of the legitimacy-money economy.

Dim. k	Variable	a_k	States ($0 \to a_k - 1$, increasing advancement)
1	Monetary regime	4	Cash, Informal credit, Formal bank, Digital
2	Legitimacy status	4	Informal, Tolerated, Recognised, Legitimate
3	Banking relationship	5	Excluded, Savings, Borrower, Full-service, Premium
4	Consumption outcome	2	Subsistence, Above-subsistence

Table 2: Birth-death calibration, utility weights, and resulting stationary moments.

k	Dimension	μ_k	ν_k	$r_k = \mu_k/\nu_k$	$\bar{z}_k^*$	α_k
1	Monetary	0.05	0.03	1.667	2.07	1.0
2	Legitimacy	0.03	0.02	1.500	2.03	2.0
3	Banking	0.04	0.03	1.333	2.78	1.5
4	Consumption	0.10	0.15	0.667	0.30	1.0

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6.2 Graph Properties

Table 3 summarises the structural parameters of $G(\Omega) \cong K_4 \,\square\, K_4 \,\square\, K_5 \,\square\, K_2$.

Table 3: Graph-theoretic properties of the legitimacy-money identity graph.

Property	Formula	Value
Identity count $\lvert\Omega\rvert$	$4 \times 4 \times 5 \times 2$	160
Degree δ	$(4-1) + (4-1) + (5-1) + (2-1)$	11
Edge count E	$\frac{1}{2} \times 160 \times 11$	880
Diameter	n	4
Spectral gap γ_{RW}	$\min(4, 4, 5, 2)/11$	$2/11 \approx 0.182$
Mixing-time bound ($\varepsilon = 0.25$)	$(11/2)\ln(160/0.25)$	≈ 36 steps

The bottleneck dimension is consumption ($a_4 = 2$), which contributes the minimum in γ_{RW}. Adding a third consumption state ($a_4 = 3$) raises γ_{RW} to $3/13 \approx 0.231$, reducing the mixing time by approximately 15%.

6.3 Analytical Welfare Benchmark

Under π^{BD}, analytical welfare is

$$W^{\mathrm{BD}} = \sum_{k=1}^{4} \alpha_k \bar{z}_k^* = 1.0 \times 2.07 + 2.0 \times 2.03 + 1.5 \times 2.78 + 1.0 \times 0.30 = 10.60.$$

6.4 Convergence of the M-H Welfare Estimator

Figure 3 plots the running-mean welfare $\hat{W}_t = t^{-1} \sum_{s=1}^{t} u(\mathbf{z}^{(s)})$ from a representative M-H run initialised at $\mathbf{z}^{(0)} = (0, 0, 0, 0)$ (most disadvantaged identity, $u(\mathbf{z}^{(0)}) = 0$).

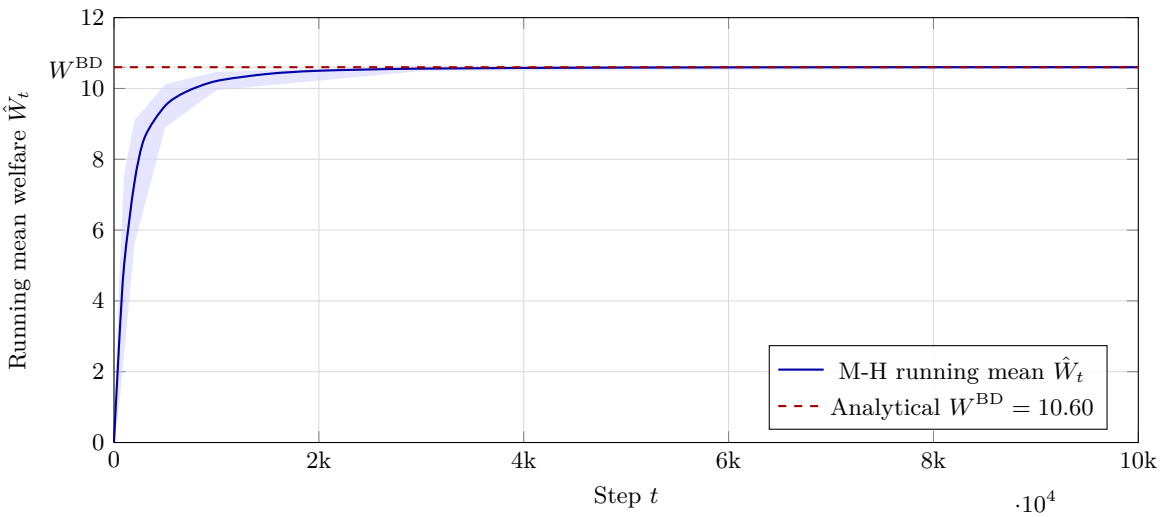

Figure 3: Convergence of the M-H running-mean welfare estimator to the analytical benchmark $W^{\mathrm{BD}} = 10.60$ (birth-death target, $\mathbf{z}^{(0)} = (0, 0, 0, 0)$, no burn-in).

The shaded band indicates approximate 95 % sampling variability over repeated runs.

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Table 4 summarises M-H performance at selected sample sizes.

Table 4: M-H welfare estimates, absolute errors, acceptance rates, and Monte Carlo standard errors at selected sample sizes ($\mathbf{z}^{(0)} = (0, 0, 0, 0)$, birth-death target, no burn-in).

Samples S	$\hat{W}_S$	$\lvert\hat{W}_S - W^{\mathrm{BD}}\rvert$	Accept rate	MCSE $\hat{\sigma}(\hat{W})$
500	9.51	1.09	0.493	0.310
1 000	10.21	0.39	0.496	0.218
2 000	10.50	0.10	0.498	0.149
5 000	10.59	0.01	0.499	0.095
10 000	10.60	< 0.005	0.500	0.067

The acceptance rate converges to ≈ 0.50, which reflects the symmetric SDP proposal and the moderate spread of π^{BD} over adjacent identities. Monte Carlo standard error decreases at the canonical $O(S^{-1/2})$ rate.

7 The Boltzmann Target and Institutional Concentration

Under $\pi^{\mathrm{Bol}}(\mathbf{z}; T)$ the identity space acts as a statistical mechanical system: institutional utility plays the role of energy and T governs the spread of probability mass. Figure 4 illustrates the birth-death chain for the banking dimension ($a_3 = 5$).

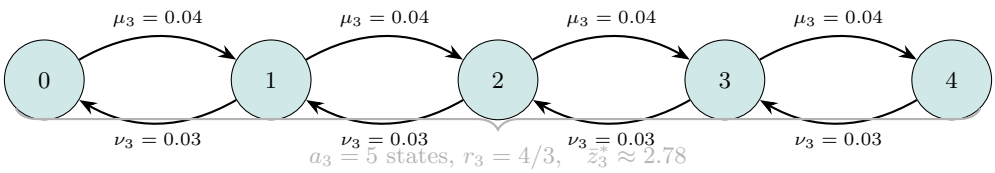

Figure 4: Birth-death chain on $a_3 = 5$ states (banking relationship, $k = 3$).

Upward rate $\mu_3 = 0.04$; downward rate $\nu_3 = 0.03$; $r_3 = 4/3 > 1$ implies upward drift toward Premium banking. Reflecting boundaries at $s = 0$ and $s = 4$.

Proposition 7.1. *π^{BD} is a Boltzmann distribution $\pi^{\mathrm{Bol}}(\cdot; T)$ for some $T > 0$ if and only if all ratios $\log r_k/\alpha_k$ are equal.*

Proof. The product form $\pi^{\mathrm{BD}}(\mathbf{z}) \propto \prod_k r_k^{z_k}$ matches $\exp(u(\mathbf{z})/T) = \exp(\sum_k \alpha_k z_k/T)$ if and only if $\log r_k = \alpha_k/T$ for all k, i.e. $T = \alpha_k/\log r_k$ is the same constant for every k. □

For the calibration of Table 2 the effective temperatures $\alpha_k/\log r_k$ are $1/\log 1.667 \approx 1.96$, $2/\log 1.500 \approx 4.93$, $1.5/\log 1.333 \approx 5.25$, and $1/\log 0.667 \approx -2.47$ respectively. Since these differ, $\pi^{\mathrm{BD}} \neq \pi^{\mathrm{Bol}}$ for any single T.

As T decreases, π^{Bol} concentrates on $\mathbf{z}^{\max} = (3, 3, 4, 1)$ with $u(\mathbf{z}^{\max}) = 3 + 6 + 6 + 1 = 16$. At $T = 1$ the highest-utility identity receives probability weight $\approx e^{16}/\mathcal{Z}(1)$, which already dwarfs the baseline of $1/160 \approx 0.006$ under the uniform distribution.

8 Discussion

Computational Complexity

Each M-H step costs $O(1)$: draw a dimension, draw a new state, evaluate the acceptance ratio. Direct computation of the stationary distribution via eigendecomposition of the $|\Omega| \times |\Omega|$ transition matrix costs $O(|\Omega|^3)$. For the 160-identity economy both methods are fast, but M-H scales polynomially as $|\Omega|$ grows—adding a fifth dimension with $a_5 = 3$ states enlarges Ω by a factor of 3 and increases δ by 2, leaving M-H equally efficient per step.

Variance Reduction

The running-mean estimator has asymptotic variance σ^2_{MH}/S, where σ^2_{MH} is the sum of the auto-covariances of the utility process along the chain [8]. Rao–Blackwellisation— averaging over all possible next states at each accepted proposal [7]—reduces this variance further at no additional sample cost.

Endogenous Dynamics

When P^* is the endogenously derived transition matrix from agent optimisation, $\pi^* = (P^*)^\top \pi^*$ has no product-form representation. M-H with the SDP proposal still applies: the acceptance ratio becomes $\pi^*(\mathbf{z}')/\pi^*(\mathbf{z})$, evaluated from the (numerical) Perron–Frobenius eigenvector of P^*. Each step remains $O(1)$ once the eigenvector is pre-computed.

Multi-Agent Fixed-Point Iteration

In the multi-agent extension [1], the equilibrium satisfies $\pi^* = \pi^* P^*(\cdot; \pi^*)$. A mean-field M-H iteration proceeds as follows: given an estimate $\hat{\pi}^{(m)}$, run Algorithm 1 with the target derived from $P^*(\cdot; \hat{\pi}^{(m)})$ to obtain a new estimate $\hat{\pi}^{(m+1)}$, then iterate. Convergence is guaranteed when the map $\pi \mapsto \pi P^*(\cdot; \pi)$ is a contraction, which can be checked via the Banach fixed-point theorem applied to the ℓ^1 norm on $\Delta(\Omega)$.

9 Conclusion

We have formalised and studied the Metropolis–Hastings algorithm for sampling institutional distributions in the Personal Economy framework. The identity graph's product structure $K_{a_1} \square \cdots \square K_{a_n}$ underpins three key results.

1. The SDP proposal is symmetric on $G(\Omega)$ (Lemma 4.2), reducing the acceptance ratio to a pure target ratio with $O(1)$ evaluation.
2. Detailed balance holds rigorously (Theorem 4.6), guaranteeing that both the birth-death product distribution and the Boltzmann distribution are invariant.
3. The M-H spectral gap satisfies $\gamma_{\mathrm{MH}} \geq \kappa\, \gamma_{\mathrm{RW}}$ (Theorem 5.1), providing an a priori mixing-time guarantee in terms of the identity graph's spectral gap $\gamma_{\mathrm{RW}} = \min_k a_k/\delta$.

In the 160-identity legitimacy-money economy the M-H running-mean welfare estimator converges to the analytical benchmark $W^{\mathrm{BD}} = 10.60$ with relative error below 0.1% after 5,000 steps and acceptance rate ≈ 0.50. The framework extends to higher-dimensional economies, endogenous transition matrices, and multi-agent fixed-point iterations, positioning M-H as a versatile computational engine for the Personal Economy research programme.

A Proof of the Factored Acceptance Ratio

Lemma A.1. *Let $\pi(\mathbf{z}) = \prod_{k=1}^n f_k(z_k)$ be any product distribution on Ω. For the SDP proposal changing only dimension k,*

$$\frac{\pi(\mathbf{z}')}{\pi(\mathbf{z})} = \frac{f_k(z_k')}{f_k(z_k)}.$$

Proof. $\pi(\mathbf{z}')/\pi(\mathbf{z}) = \prod_j f_j(z_j')/\prod_j f_j(z_j)$. Since $z_j' = z_j$ for all $j \neq k$, every factor with $j \neq k$ contributes 1, leaving only $f_k(z_k')/f_k(z_k)$. □

Proposition 4.3 follows with $f_k(s) = r_k^s/(\sum_j r_k^j)$; Proposition 4.4 with $f_k(s) = \exp(\alpha_k s/T)/(\sum_j \exp(\alpha_k j/T))$.

B Stationary Distribution of the Birth-Death Chain

Let $\{X_t\}$ be a birth-death chain on $\mathcal{S} = \{0, \ldots, m-1\}$ with $p(s, s+1) = \mu$ for $s < m-1$, $p(s, s-1) = \nu$ for $s > 0$, and reflecting boundaries. Setting $r = \mu/\nu$:

$$\pi^*(s) = \begin{cases} \dfrac{(1-r)\, r^s}{1-r^m} & r \neq 1, \\ \dfrac{1}{m} & r = 1. \end{cases} \tag{13}$$

The mean is

$$\bar{z}^* = \sum_{s=0}^{m-1} s\, \pi^*(s) = \begin{cases} \dfrac{r}{1-r} - \dfrac{m r^m}{1-r^m} & r \neq 1, \\ \dfrac{m-1}{2} & r = 1. \end{cases}$$

For the consumption dimension ($m = 2$, $\mu = 0.10$, $\nu = 0.15$, $r = 2/3$):

$$\pi^*(0) = \tfrac{3}{5} = 0.60, \quad \pi^*(1) = \tfrac{2}{5} = 0.40, \quad \bar{z}_4^* = 0.40.$$

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Glossary of Notation

n Number of institutional dimensions.

Z_k Finite state set of dimension k; $|Z_k| = a_k \geq 2$.

Ω Identity space $Z_1 \times \cdots \times Z_n$; $|\Omega| = \prod_k a_k$.

$\mathbf{z} = (z_1, \ldots, z_n)$ An economic identity; element of Ω.

$G(\Omega)$ Identity graph: vertices Ω, edges between identities differing in exactly one coordinate.

$\square$ Graph Cartesian product.

K_m Complete graph on m vertices.

δ Common vertex degree in $G(\Omega)$: $\delta = \sum_k (a_k - 1)$.

γ_{RW} Spectral gap of the uniform random walk on $G(\Omega)$: $\gamma_{\mathrm{RW}} = \min_k a_k/\delta$.

$u(\mathbf{z})$ Institutional utility: $u(\mathbf{z}) = \sum_k \alpha_k z_k$.

α_k Utility weight for dimension k.

μ_k, ν_k Upward and downward birth-death rates for dimension k.

r_k Birth-death ratio: $r_k = \mu_k/\nu_k$.

$\pi_k^*(s)$ Marginal birth-death stationary for dimension k: $\pi_k^*(s) = r_k^s / \sum_j r_k^j$.

π^{BD} Birth-death product distribution: $\pi^{\mathrm{BD}}(\mathbf{z}) = \prod_k \pi_k^*(z_k)$.

π^{Bol} Boltzmann distribution: $\pi^{\mathrm{Bol}}(\mathbf{z}) \propto \exp(u(\mathbf{z})/T)$.

T Temperature parameter of the Boltzmann distribution.

$\mathcal{Z}(T)$ Partition function: $\mathcal{Z}(T) = \sum_{\mathbf{z}} \exp(u(\mathbf{z})/T)$.

$q(\mathbf{z}, \mathbf{z}')$ SDP proposal kernel: $q(\mathbf{z}, \mathbf{z}') = 1/(n(a_k - 1))$ if $\mathbf{z} \sim \mathbf{z}'$ in $G(\Omega)$, else 0.

$\alpha(\mathbf{z}, \mathbf{z}')$ M-H acceptance probability: $\alpha(\mathbf{z}, \mathbf{z}') = \min(1, \pi(\mathbf{z}')/\pi(\mathbf{z}))$.

P_{MH} Transition matrix of the M-H chain.

γ_{MH} Spectral gap of the M-H chain; bounded below by $\kappa\,\gamma_{\mathrm{RW}}$.

κ Target concentration ratio $\kappa = \pi_{\min}/\pi_{\max}$.

$t_{\mathrm{mix}}(\varepsilon)$ ε-mixing time: smallest t such that total variation distance from stationarity is $\leq \varepsilon$.

$W(\pi)$ Expected social welfare under π: $W(\pi) = \sum_{\mathbf{z}} \pi(\mathbf{z})\, u(\mathbf{z})$.

W^{BD} Analytical welfare under π^{BD}; equals 10.60 in the case study.

$\hat{W}_S$ M-H running-mean welfare estimator after S steps: $\hat{W}_S = S^{-1} \sum_{t=1}^{S} u(\mathbf{z}^{(t)})$.

MCSE Monte Carlo standard error: $\hat{\sigma}(\hat{W}) = \hat{\sigma}_{\mathrm{MH}}/\sqrt{S}$.

The End